

Senior Design Fall 2025 Debrief

Hello! If your reading this then you have been given this wheelchair probably by Dr. Wei as a Senior Design Project for the future! My Name is Evan Rees and this was the first year we utilized and repaired this chair, so here are some lessons we learned that you can use in your iteration of a project!

Before I go into it too much, here is my contact information:
Email: ev596984@ucf.edu / evan@thereesgroup.com
Phone: +1 (941)-465-8774

If you every need some help getting antiquated or have a question that isn't too easy to solve, shoot me a text/email and I will get back to you if I have a solution. No promises as I'm a student just like you and made a ton of mistakes and this chair still has a few unknowns. If we had more than two semesters then we maybe would have figured them all out ;]

SMART-STRIDE

This was our project for Summer/Fall 2025, you can access our website at:
<http://maverick.eecs.ucf.edu/seniordesign/su2025fa2025/g11/>
or if that's not your fancy we have a GitHub here:
<https://github.com/lavabrothers/SelfDrivingWheelchairSD2>

Long story short we designed the wheelchair to move wireless and somewhat autonomously powered by a Pi 5, a 4 output DAC, and a Kinectv2 Sensor. All the code is made with these elements in mind, with a ESP32 as a wireless controller. It contains 4 modes:

- Wireless Manual Control Mode
 - This mode basically just sends joystick inputs to the wheelchair, as if it's plugged in directly
- Cruise Control Mode
 - This drives forward and will stop if it detects anything too close such an an object or a wall
- Follow Mode
 - This will identify the closest person to the wheelchair, and attempt to maintain distance from the person and keep them centered in it's vision
- Map Mode
 - This will spin the wheelchair 360 degrees and construct a depthmap of the environment around it



QR Code: Website



QR Code: GitHub

The GitHub has the most up to date info, but the website can get you an idea on how we progressed over the Senior Design Class. I recommend if you setup something similar, go to commit 22fafa5 as that has a bunch of crummy dev scripts you can use to test and experiment. However the repo is built in a way to do most everything already.

At time of writing I'm not sure if Dr. Wei will have you start your own project, or jump off the work we have already done, but if it's the latter I have some ideas for you:

IDEAS

- Utilize the unused map mode to stitch together larger environments
- Fix follow mode to be more reliable and follow a single person without being stolen by another close person
- Assisted Manual Control that will help the user avoid obstacles while in manual mode
- Utilizing a rejected voice control mode (read back in the code commits), enable voice controls
- Integrate AI of some kind for natural voice commands
 - Wei mentioned this to use when we initially were in development, but we ended up focusing on basic functionality
- (Very Hard) Create a Full Self Driving suite with the ability for a user to select a point in space and have it travel there

There! I hope those give some ideas on what to do, if not, good luck out there!

FIRST STEPS

Right now the wheelchair is not functional in it's current form, that's because you will need to give it two 12V Lead Acid batteries in series to tap into to power it. For testing you can use any 12V batteries, but probably a good idea to make them the same. If you want to get batteries that are perfect for the wheelchair, get 22NF class batteries.

For your convenience the levers in the back near the motors can be turned to allow for it to "free wheel" so you can push the wheelchair when it is powered.

MAKE SURE YOUR NOT IN FREE WHEEL BEFORE POWERING ON THE CHAIR.



QR Code: Old User Manual

Refer to the physical manual there or digitized manual you can read. It'll give you a basic idea on how to utilize the wheelchair in it's original *intended* use. But if you want to use it differently you can use a similar method that we did.

Brain Slugging the Chair

While you could try to tap into the motors yourself, you should instead utilize the 1999 electronics inside to your advantage. Look over `wheelchair_control.py` we use to get an idea of how it works, but basically the joystick port can be manipulated to have the wheelchair move without needing to do anything special to the wheelchair's electronics. The joystick port is located next to the blue potentiometer on the controller PCB. The order of the pins are as follows:

TOP

- Left Channel (WHITE)
- Right Channel (BLUE)
- Backwards Channel (BROWN)
- Forward Channel (YELLOW)
- Ground (BLACK)
- 5V Rail (RED)

BOTTOM

If you hook into the channel pins you first need to make sure each pin is 2.5V when booting up the chair, or else it won't boot. And you must keep a ratio between the Forward/Backward and Left/Right Channels. So Forward Channel can be 3V and the Backward Channel can be 2V to move Forward.

Congrats, you can do whatever you want to the wheelchair. Feel free to refer to our Github to get any more insight and get access to our hardware schematics made in Fusion360. I highly recommend you test the wheelchair with fresh batteries to get an idea if it's functional. Here is some roadblocks we hit that you should be aware of if you plan to follow in our footsteps

CAUTIONARY

- Since we need a 5V signal to the DAC, if your using I2C like we did with a Pi, you must use a Level Converter to ramp up the 3.3V I2C lines on the Pi to be accepted by the DAC
- The Kinectv2 is a computationally heavy device and requires USB 3.0. We overclocked our Pi 5 to 2.8Ghz and increased the USB buffer to a Gig. If you convert our python to C you maybe able to get more headroom, but consider accelerators or other SBC if you plan to do more on board the device.
 - Alternatively design your 5V regulator to have a higher voltage to facilitate higher over clocks
- We used MobileNET-SSD, you may want to get an accelerator or get GPU acceleration working in your deployment
- If you fry the Pi, you will fry anything connected via the 5V rail like USB devices, do not fry your stuff. Trust me
- Get ahead of deadlines as this project is imense and hard to transport. The more you work eaely on the better.
- The Kinect sensor uses IR for depth, it maybe a bit spotty if there are IR night vision cameras so make sure you account for the interference like we tried last minute